Higher-MI Predictor

Does information sharing among the output scores warn us before grokking?

A visual guide to Predictor 7 of 8 · *A Unified Benchmark of Grokking Predictors in Neural Networks*

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The answer first

Verdict: Higher-MI is CLOSED, negative, under the frozen evaluation protocol.
Its event (the global maximum of population O-information) came *after* the grok epoch in **all five seeds**.
Frozen-protocol scores: lead fraction $= 0$ · Spearman $\rho = 0.7$ · mean relative lead $= -0.4254$.
All three conditions fail.

1. What was Higher-MI supposed to detect?
It was supposed to give an early warning that grokking is coming. The idea comes from Pomarico *et al.* (2025). They reported that a redundancy peak in the output scores appears at the grokking transition.

2. What did we actually measure?
At every saved checkpoint, we ran the model on the 8,939 unseen test examples. From each example's 113 output scores, we kept three summaries: the correct-answer score, the strongest wrong score and the background score. We then measured how these three quantities share information across the test population. That measure is called *population O-information*, $\Omega_{\text{pop}}(t)$.

3. What do the graphs show?
$\Omega_{\text{pop}}(t)$ is not a clean "rise, peak, settle" curve. It spikes up and falls back again and again, for the whole of training. The first spike appears at epoch 1,900 in every seed, even though the seeds grok at very different times. The grok epoch falls in the middle of this pattern. It does not mark a visible change.

4. Did Higher-MI predict grokking in advance?
No. In every seed the largest spike came after grokking, by 1,579 to 11,090 epochs. Not a single seed gave an early warning.

5. What should I conclude?
This particular adaptation of O-information does not work as an early-warning signal in this benchmark. That is the full claim. It does *not* show that O-information is useless. It does *not* show that the original paper is wrong. The paper studied a different model, a different task, and claimed only that the peak *coincides* with grokking.

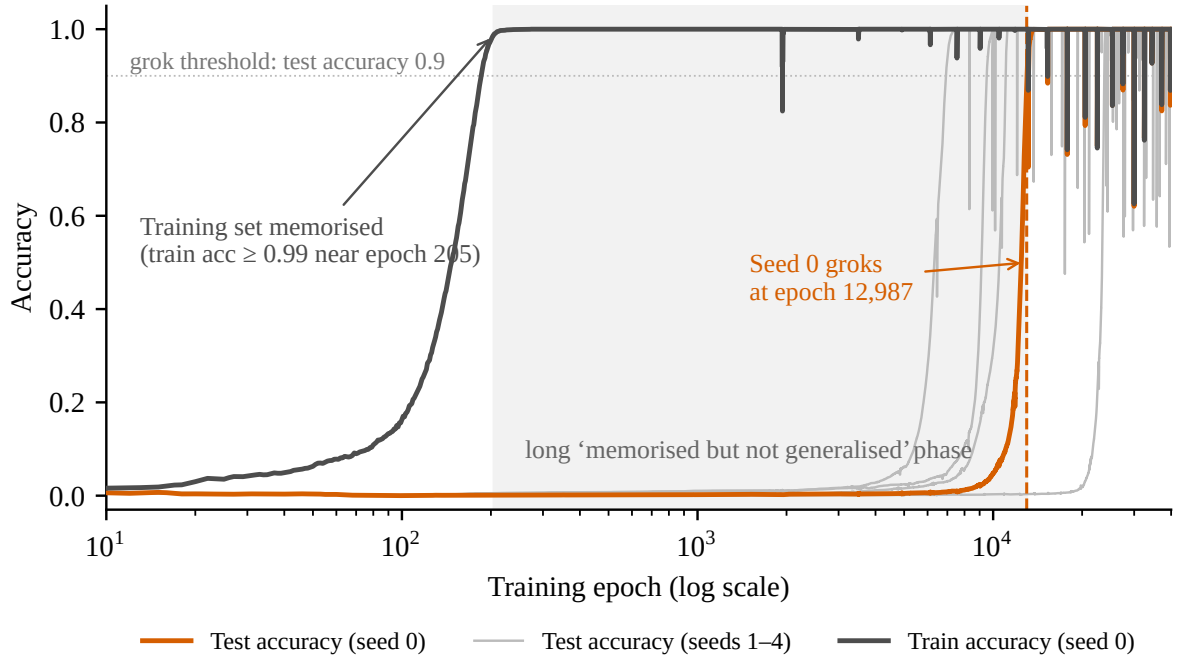
How to read this guide. The story runs in six steps: the problem (§1) → the idea (§2–3) → how the measurement works (§4) → what the graphs reveal (§5) → how the prediction was judged (§6) → what we learned (§7). §8 is viva preparation. Technical details are in the Appendix. Colour is used consistently: **orange** = grokking, **blue** = the predictor's event, grey = background. Line styles and markers also differ, so the graphs read without colour.

1 The problem: grokking

Intuition. Think of a student who memorises every answer in the practice book. On the practice book, the student scores 100%. On a fresh exam paper, the student still fails. Then, after a long time of further practice, the student suddenly understands the method and starts passing new papers. That sudden late understanding is grokking.

The task. The model learns modular addition, $c = (a + b) \bmod 113$, with $a, b \in \{0, \ldots, 112\}$. There are $113 \times 113 = 12{,}769$ pairs. 30% of them (3,830 pairs) are used for training. The remaining 8,939 pairs form the test set. The model never sees them during training. The model is the standard Nanda *et al.* (2023) setup: a one-layer transformer, trained full-batch with AdamW and weight decay 1.0 for 40,000 epochs. Five seeds (0–4) are trained. A seed fixes both the initial weights and the train/test split.

Graph 1. When does the model memorise, and when does it generalise?



Training and test accuracy from the saved training histories. The x-axis is logarithmic, so the early memorisation phase and the late generalisation are both visible. Upward movement means the model gets more examples right.

What should I notice? (observation) Training accuracy reaches 0.99 at about epoch 205 in all five seeds. Test accuracy stays near zero for thousands of epochs after that. It then rises sharply. For seed 0 it crosses 0.9 at epoch 12,987.

What does this tell us? (interpretation) The model memorises long before it generalises. We define the **grok epoch** G as the first epoch at which test accuracy exceeds 0.9. The five grok epochs differ widely: $G = 12{,}987,\ 6{,}921,\ 9{,}510,\ 11{,}019,\ 23{,}747$ for seeds 0–4. This spread is useful. A genuine predictor must fire early in *each* seed, and it must fire later for seeds that grok later.

The research question. Can some pattern in the model's output scores warn us that grokking is approaching? A quantity that moves only at grokking, or after it, cannot serve as a warning.

2 The idea: three characters in every decision

Intuition. For each input, the model gives one raw score, called a **logit**, to each of the 113 possible answers (classes). A higher logit means more belief in that class. A logit is not a probability; it can be any real number. The predicted answer is the class with the largest logit.

Instead of tracking all 113 logits, Higher-MI follows three "characters" who compete to describe the decision:

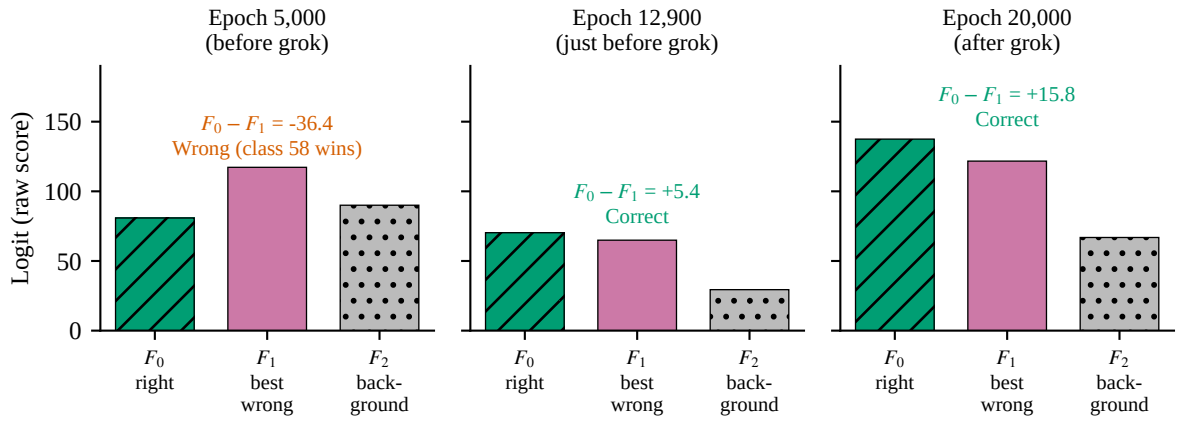
- F_0, the **correct-answer score**: how strongly does the model score the right answer?
- F_1, the **strongest-wrong-answer score**: how strong is the most convincing incorrect answer?
- F_2, the **background score**: the average score of the remaining 111 incorrect answers.

Technical meaning. For a test example with correct class c and logits $\ell[0], \ldots, \ell[112]$:

$$F_0 = \ell[c], \qquad F_1 = \max_{j \neq c} \ell[j], \qquad F_2 = \frac{1}{111} \sum_{j \neq c,\ j \neq r} \ell[j],$$

where r is the runner-up class, the wrong class that achieves F_1. The model answers correctly exactly when $F_0 > F_1$. The gap $F_0 - F_1$, called the **margin**, says how decisive that choice is.

Graph 4. One unseen example, (23, 45) → 68, seed 0: how the three scores change



One test example, $(23, 45) \to 68$, seed 0, computed from the saved checkpoints. Bar height is the logit. The label above each panel gives the margin $F_0 - F_1$. *Illustrative single example only; the predictor itself uses all 8,939 test examples.*

What should I notice? (observation) At epoch 5,000 the wrong class 58 scores far above the right class. Even the background F_2 sits above F_0. At epoch 12,900, just before seed 0 groks, the right class has moved to the top and the background has dropped far below. At epoch 20,000 the margin has grown further.

What does this tell us? (interpretation) At epoch 5,000 the model has memorised its training set, but for this unseen pair it has not learned the rule. Later it has. One example only shows the mechanics, though. It cannot tell us about the whole model.

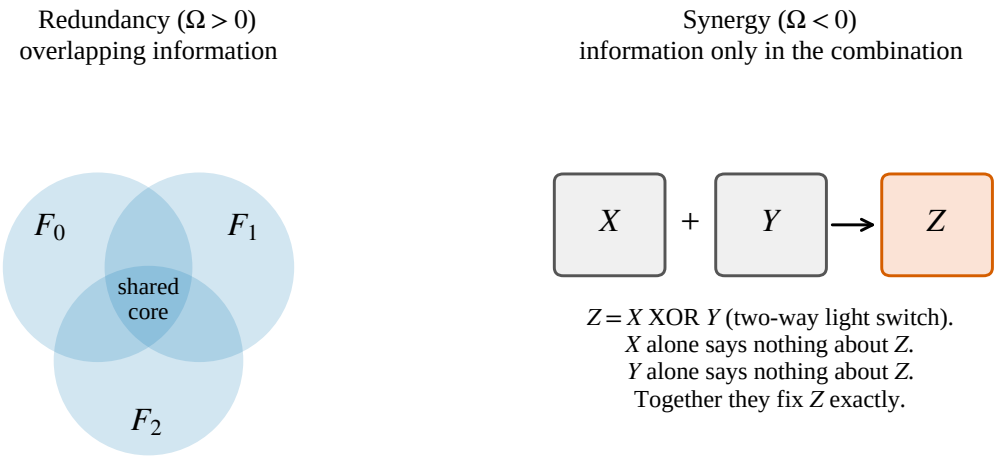
From one example to the population. At one checkpoint, every test example gives one triplet (F_0, F_1, F_2). So we get 8,939 triplets, a cloud of points in three dimensions. We treat F_0, F_1 and F_2 as three random

variables. This cloud is their sample. The fraction of points with $F_0 > F_1$ is simply the test accuracy.

Why reduce 113 logits to three? O-information, in its basic form, is defined for three variables. The source paper had a three-class problem, so it had exactly three scores. Our task has 113 classes. The (F_0, F_1, F_2) reduction is a *project-specific adaptation*. It is not in the paper. The never-trained output for the "=" token is removed before the reduction, so it can never act as a false runner-up.

3 What O-information means

Intuition. Three variables can relate in two opposite ways. **Redundancy**: they repeat the same information. Three newspapers reporting the same cricket score are redundant. Reading a third paper tells you nothing new. **Synergy**: information appears only when variables are combined. A staircase light with two switches is the classic case. The bulb is on when the switches differ. One switch alone tells you nothing about the bulb. Both together tell you everything.



Schematic. Left: overlapping information (redundancy). Right: information that exists only jointly (synergy, the XOR case).

Technical meaning. Mutual information $I(F_0; F_1)$ is what F_0 and F_1 share. Conditional mutual information $I(F_0; F_1 \mid F_2)$ is what they still share once F_2 is known. For three variables, O-information is

$$\Omega(F_0, F_1, F_2) = I(F_0; F_1) - I(F_0; F_1 \mid F_2).$$

- $\Omega > 0$: knowing F_2 *removes* part of what F_0 and F_1 share. The three overlap. This is redundancy.
- $\Omega < 0$: knowing F_2 *increases* what F_0 and F_1 share. Some information lives only in the triple. This is synergy.

In theory Ω is symmetric in its three arguments. The finite-sample estimate is not exactly symmetric. So the code computes Ω for all $3! = 6$ orderings and reports their mean. Units are nats.

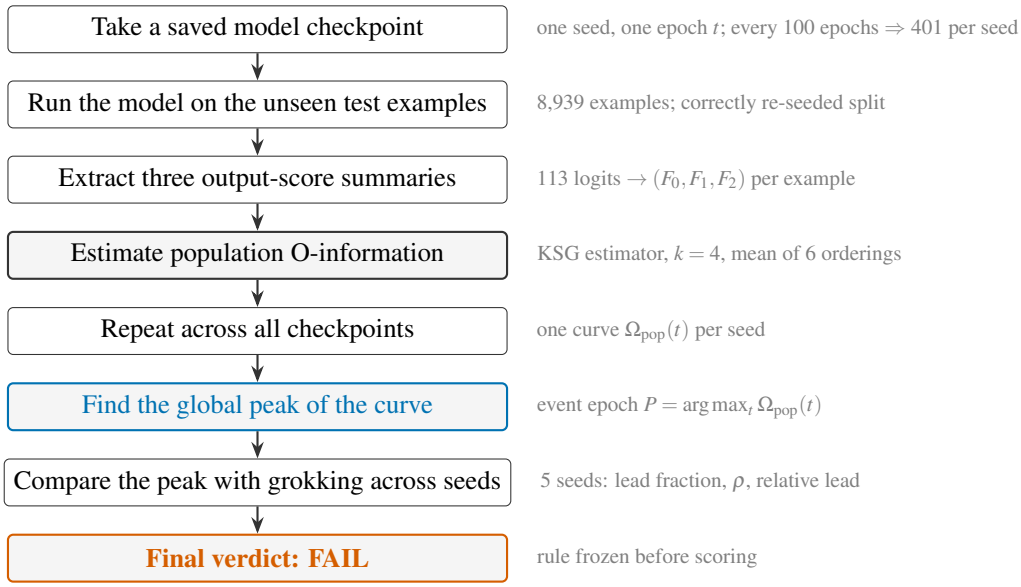
Implication. The sign tells us the *kind* of structure, not the model's state. A positive value does not mean the model has grokked. A negative value does not mean it has not. A large shared information also does not imply a large Ω: at epoch 20,000, F_0 and F_1 are almost perfectly correlated across examples, yet Ω is only about 0.36 nats, because Ω counts only the part that F_2 explains away.

The hypothesis. The paper's story is: synergy ($\Omega < 0$) at random initialisation, a switch to redundancy as the model learns, and a redundancy peak *at* grokking. This thesis tests a stronger claim:

> *The peak of $\Omega_{\text{pop}}(t)$ comes before the grok epoch, with a timing that follows the grok epoch across seeds.*

4 How the measurement works

The whole experiment fits in one picture. Every box uses only saved checkpoints; no model was re-trained.



The Higher-MI pipeline, from checkpoint to verdict.

Why held-out (test) examples? Grokking is about generalisation. On training examples the model is confident almost from the start, because it has memorised them. Only the unseen examples carry the change we want to detect.

How do we estimate information from 8,939 points? F_0, F_1, F_2 are continuous. We do not know their probability densities. Cutting the axes into bins would make the answer depend on an arbitrary bin width. The **KSG estimator** (Kraskov–Stögbauer–Grassberger) avoids bins. For each point, it finds the distance ε_i to its k-th nearest neighbour in the joint space. It then counts how many points fall strictly within ε_i along each single axis. If two variables are strongly dependent, a small joint neighbourhood still contains few marginal neighbours, so the counts are small and the estimate is large. The only parameter is k; the project uses $k = 4$. The formulas are in the Appendix.

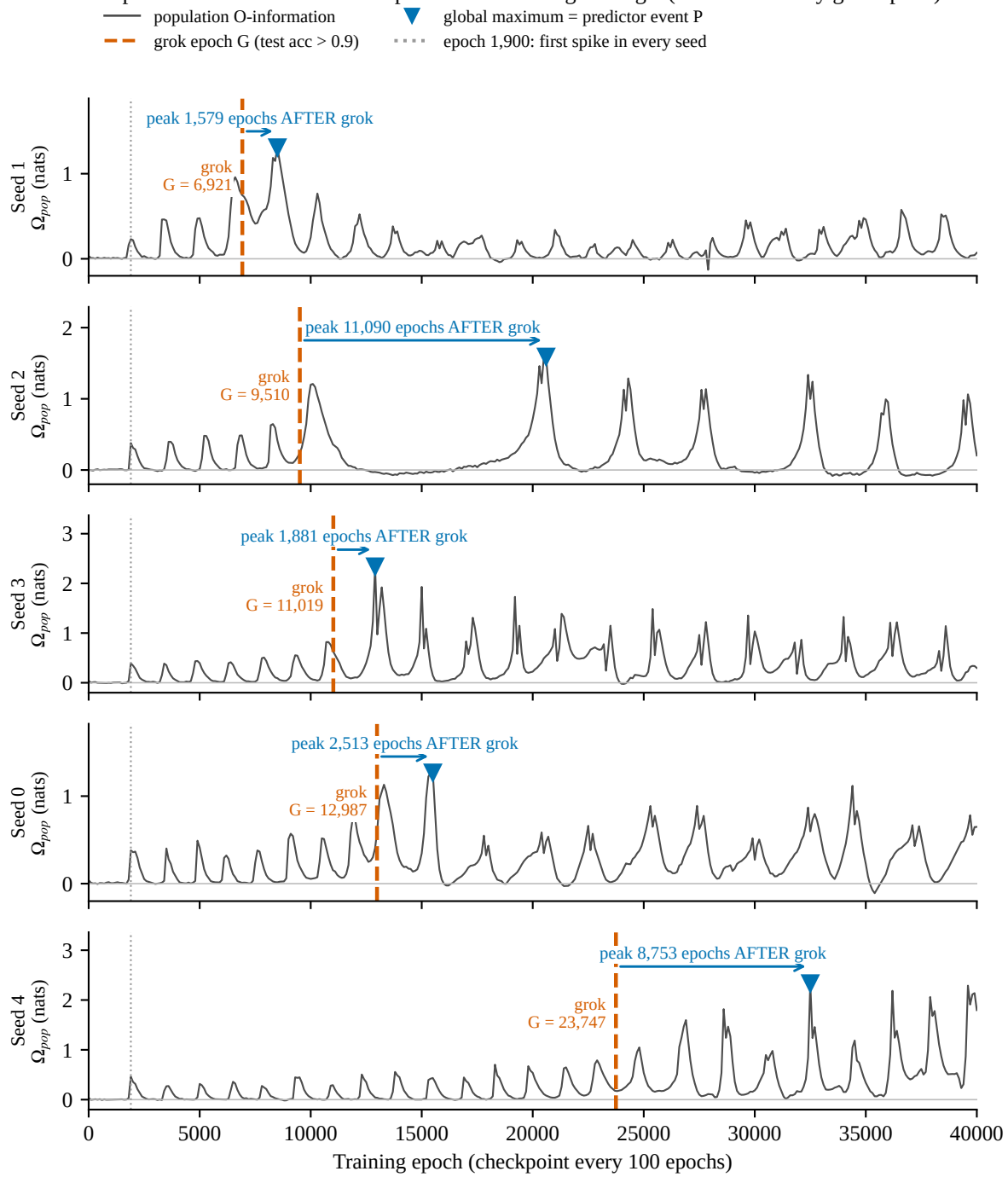
5 What the graphs reveal

The key graph: O-information against grokking

What should I notice? (observation)

- **No strongly negative start.** At epoch 0, Ω_{pop} lies between $+0.002$ and $+0.040$ in every seed. Negative values are rare and small. The lowest is -0.130.

Graph 2. Does the O-information peak arrive before grokking? (seeds ordered by grok epoch)



Saved $\Omega_{\text{pop}}(t)$ curves for all five seeds, at all 401 checkpoints, on their own y-scales. Upward movement means more redundancy among F_0, F_1, F_2 across the test population. Each axis is drawn with its true range; no smoothing.

- **Repeated spikes for the whole run.** The curve rises and falls back many times, before and after grokking.
- **A common clock.** In all five seeds, Ω_{pop} first exceeds 0.2 at exactly epoch 1,900. Before grokking, spikes then recur roughly every 1,400–1,600 epochs.
- **No landmark at grok.** The orange dashed line lands in the middle of the oscillation.
- **The highest spike (blue triangle) is always to the right of the orange line.**

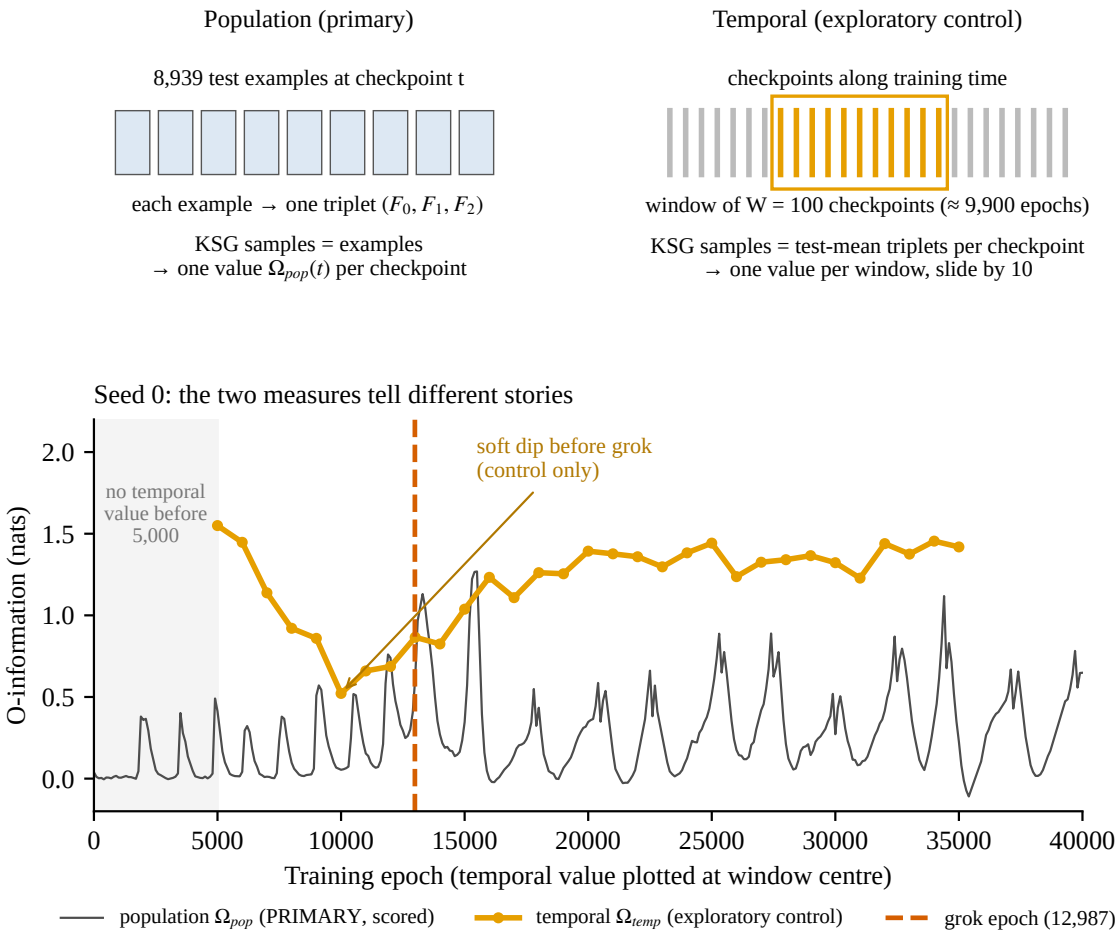
- **Amplitude drift differs by seed.** Seed 4 grows from about 0.04 to 1.80. Seed 1 ends almost where it started (0.033 to 0.070).

What does this tell us? (interpretation) A clock that ticks at the same epoch in all seeds cannot be tracking grokking, because grokking happens at very different epochs in different seeds. So Ω_{pop} seems to follow general training dynamics, such as how logit margins develop, rather than the grokking transition. The paper's sign story (strongly negative at initialisation, then positive) is not reproduced here. The cause of the epoch-1,900 onset was not investigated. It remains an open question.

Population versus temporal O-information

The paper and this project use the word "samples" differently. This is the biggest adaptation in the predictor.

Graph 5. Population versus temporal O-information



Top: what one KSG sample means in each measure. Bottom: both saved curves for seed 0 on a shared time axis. The temporal value is plotted at the centre of its window.

The two measures. In the paper, one training step gives one sample; O-information there describes how the three scores *move together over training time*. Our primary measure, $\Omega_{\text{pop}}(t)$, instead uses the 8,939 test examples at one frozen checkpoint. It describes *structure across examples at one moment*. It has many samples and no window parameter. But it measures a different thing. As a check, an exploratory

temporal control Ω_{temp} was also computed in the paper's spirit. It takes the test-mean of F_0, F_1, F_2 at each checkpoint and slides a window of 100 checkpoints (about 9,900 epochs), stride 10. That gives 31 values, centred from epoch 5,000 to 35,000.

Why was the control needed? All three numbers come from the same forward pass on the same input. So across the population they may share a common cause: how easy or hard the example is. An easy example has high F_0 and very low F_2. A confusing one has F_0 and F_1 close together. Such shared difficulty creates redundancy by itself, whether or not the model has grokked.

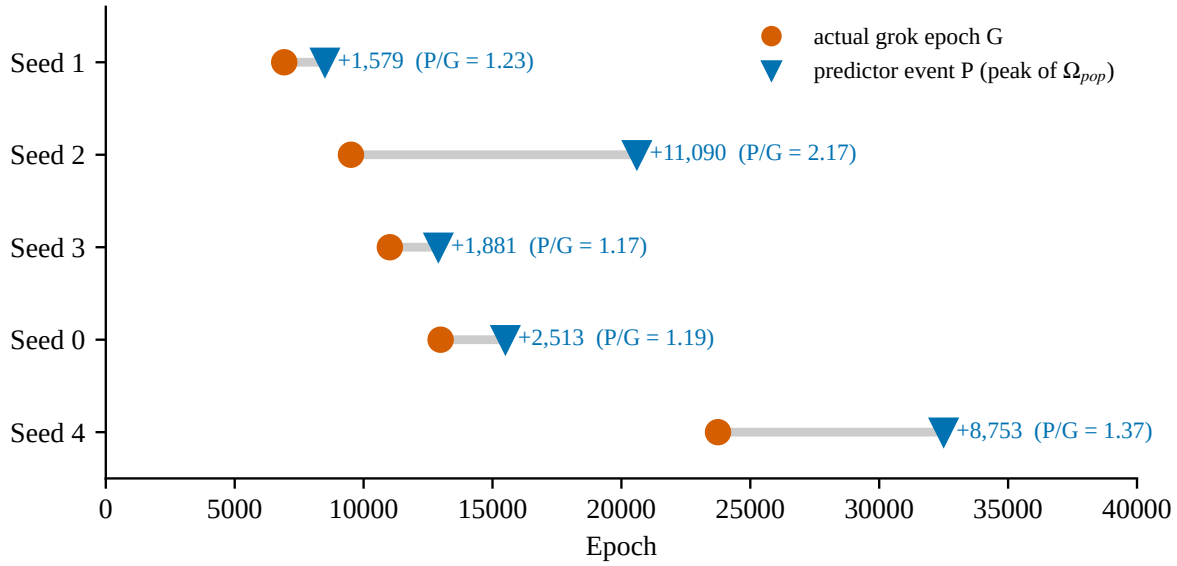
What should I notice? (observation) The two curves have different shapes. The temporal control is smooth. It starts near 1.55, dips to about 0.52 near epoch 10,000, just before grok, then stays around 1.2–1.45. It is never negative in any seed. The population curve shows no such dip.

What does this tell us? (interpretation) Because the shapes differ, the population signal is not just a copy of one "shared difficulty" factor. That partly answers the confound worry; it does not remove it. The two measures are *not interchangeable*. The soft dip belongs to the control, not to the scored signal, and the control's coarse windows cannot place it precisely. So the dip is a curiosity, not a rescued predictor.

6 How the prediction was judged

From a curve to one epoch. The protocol compares one predictor epoch with one grok epoch per seed. So a rule must pick one epoch from each curve, and it must be fixed *before* seeing the score. The primary rule, frozen on 3 October 2026, is the **global maximum**: $P = \arg\max_t \Omega_{\text{pop}}(t)$. Two reasons: it is the paper's own landmark (the redundancy peak), and it always exists, so every seed gets an event. It also matches the other predictors, whose primary rules are all global extrema.

Graph 3. Predicted event versus actual grokking, seed by seed



Grok epoch G (orange circle) and predictor event P (blue triangle) for each seed. $P/G > 1$ means the event came after grokking.

What should I notice? (observation) An early warning would place the blue triangle *left* of the orange circle. A simultaneous event would place them on top of each other. Here the triangle is to the right in

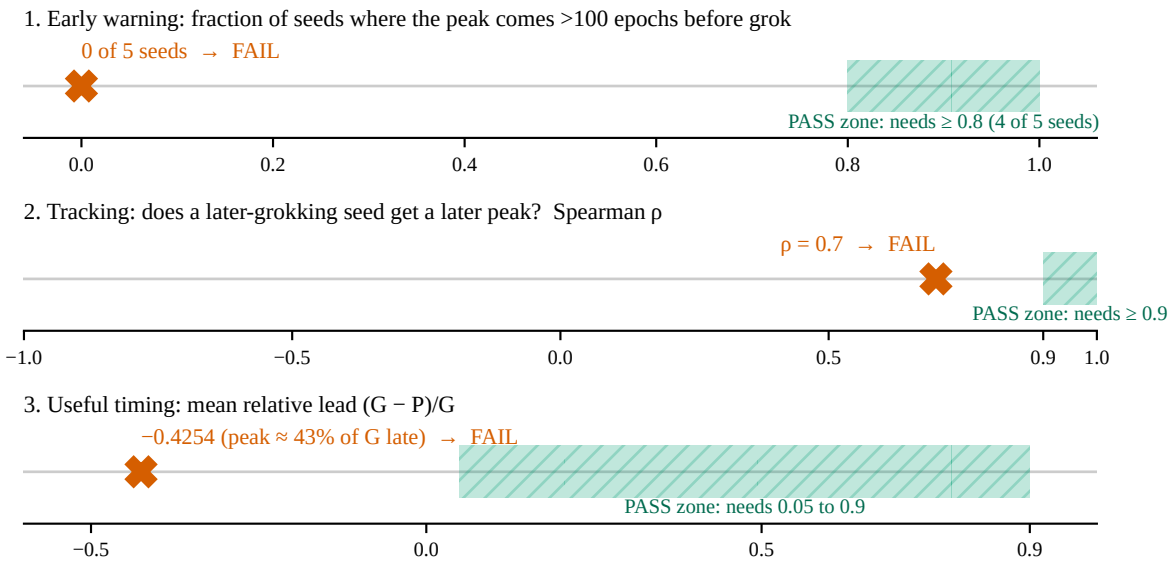
all five seeds. The event lands 1.17 to 2.17 times the grok epoch. Seed 1 is earliest in both G and P, and seed 4 is latest in both. Seeds 2 and 3 swap places.

What does this tell us? (interpretation) The signal is a late signal everywhere. It partly follows the seed-to-seed order of grokking, but it never comes first.

The frozen protocol. The same three-part rule is used for every predictor in the thesis. All three must pass:

1. **Early warning.** A seed counts as a lead if $P < G - 100$. The *lead fraction* must be at least 0.8 (four of five seeds).
2. **Tracking.** Spearman rank correlation ρ between G and P across seeds must be at least 0.9. This threshold was calibrated, not chosen by hand: it is the 99th percentile of ρ for 10,000 purely random predictors.
3. **Useful timing.** The mean relative lead, the average of $(G-P)/G$, must lie between 0.05 and 0.9. Below 0.05 the warning is too late to use. Above 0.9 it fires almost at initialisation and carries no timing information.

Graph 6. The frozen-protocol scorecard: all three conditions must pass



Each row is one condition. The hatched green band is the pass zone. The cross is the actual Higher-MI value.

What should I notice? (observation) All three crosses lie outside their pass zones.

What does this tell us? (interpretation) The failure is not borderline. The first and third conditions fail because the peak comes *after* grok; a negative relative lead means "late". On average the peak lands about 43% of a grok-time after grokking. The second condition would fail even on its own: $\rho = 0.7$ is below 0.9.

7 What we learned

The conclusion, stated carefully

Observed. Population O-information on the logit triplet peaks after grokking in 5 of 5 seeds. It oscillates throughout training, with a seed-independent onset at epoch 1,900. **Interpreted.** It appears to track general training dynamics, not the grokking transition. **Status.** Higher-MI is closed as negative. It is the seventh predictor to close negative, after L2 Norm, Dropout, Spectral, AGE, HTSR Alpha and Weight-PCA.

Why this does not contradict the original paper

	Pomarico *et al.* (2025)	**This project**
Model	Matrix product state classifier	One-layer transformer
Task	Three-class problems (reduced Fashion-MNIST; hyperspectral land cover)	Modular addition, 113 classes
Variables	The three class scores	Project-defined (F_0, F_1, F_2)
Samples	Training steps (temporal)	Test examples at one checkpoint
Claim tested	Peak *coincides* with grokking	Peak *leads* grokking

So the paper never made a "leads" claim. Our stricter test shows that the signal does not become a leading indicator when transferred to the standard grokking transformer. Here the peak is not even coincident.

Can we trust the pipeline?

Two silent bugs were found and fixed before scoring. Neither would have caused a crash.

- **Inclusive versus strict neighbour counting.** KSG requires counting points *strictly* inside ε_i. The library call counted "$\leq \varepsilon_i$", which wrongly includes the neighbour that defines ε_i. This affected about half the points in one cross-check, and changed Ω by about 2.6 times in one test case. The fix queries with the largest float strictly below ε_i. It then matched a brute-force reference bit for bit.
- **Test split not re-seeded.** The recompute path built the test set without first setting the seed. So many "test" examples were really training examples. That would bias exactly the early checkpoints a warning signal needs. The fix rebuilds the split with the saved seed. The rebuilt seed-0 split has 8,939 examples and reproduces the stored Ω_{pop} values exactly.

The estimator also passes known-answer tests: independent Gaussians give $I \approx 0$; near-copies give $\Omega > 0$; a product-type construction gives $\Omega < 0$; a hand-built fake model gives the right triplet and ignores the "=" class. And the protocol itself passes an ideal synthetic predictor 100% of the time.

What this establishes, and what it does not. These checks make the *numbers* reliable. They do not make the predictor scientifically useful. A correct measurement of a signal that does not lead is still a signal that does not lead.

Assumptions and limitations

- **Adaptation choices.** The (F_0, F_1, F_2) reduction, the population reading of "samples", $k = 4$ (stated in the paper only for transfer entropy), and the global-maximum rule. A different reduction could give a different signal.
- **Only five seeds.** With $n = 5$, Spearman ρ can take only 11 values. A random predictor that always fires before grok can pass the protocol about 22% of the time (synthetic control E2). This caveat does not affect Higher-MI, which fails on lead, not on ρ alone.
- **Checkpoint spacing of 100 epochs.** Any feature narrower than 100 epochs is invisible.
- **Coarse temporal control.** One window spans about 9,900 epochs, and the first value is at epoch 5,000.
- **Shared-example-difficulty confound.** Partly addressed by the temporal control, not fully removed.
- **The epoch-1,900 onset.** Unexplained. This question is still open; it was not investigated.

To investigate the signal further, one would need: more seeds; denser checkpoints around epoch 1,900 and around each grok epoch; a test of what drives the common onset (for example, whether it coincides with a change in loss or in logit scale); a control that removes example difficulty (for example, conditioning on per-example margin); and alternative reductions of the 113 logits.

8 What I must be able to explain

1. **What is grokking?** A model fits its training set early but generalises only much later. Here, train accuracy hits 0.99 near epoch 205; test accuracy crosses 0.9 between epochs 6,921 and 23,747, depending on the seed.
2. **What does Higher-MI try to detect?** Whether the way the output scores share information changes *before* grokking, so that it can warn us in advance.
3. **Why three logit summaries?** O-information is defined for three variables. The paper had three classes; we have 113. So we keep the three quantities that decide the prediction: right answer, strongest wrong answer, background. This reduction is our adaptation.
4. **What are redundancy and synergy?** Redundancy: variables repeat shared information, $\Omega > 0$. Synergy: information exists only jointly, as in XOR, $\Omega < 0$. $\Omega = I(F_0; F_1) - I(F_0; F_1 \mid F_2)$.
5. **Why held-out examples?** Grokking is a change in generalisation. Training examples are memorised early and hide that change. Using training examples by mistake was exactly Bug 2.
6. **Population versus temporal O-information?** Population: samples are test examples at one checkpoint, giving structure at one moment. Temporal: samples are checkpoints in a time window, giving co-movement over training. They had clearly different shapes here, so they are not interchangeable.
7. **Why the global maximum?** It is the paper's landmark, it always exists, it matches the other predictors' rules, and it was frozen before scoring so it could not be tuned to pass.
8. **Why did the predictor fail?** The peak came after grok in all five seeds. Lead fraction 0, $\rho = 0.7$, mean relative lead -0.4254. All three conditions fail.
9. **Does this show the paper is wrong?** No. Different model, task, variables and meaning of "samples". The paper claimed coincidence; we tested lead.
10. **What do the bug fixes establish?** That the numbers are correct: the estimator matches a brute-force reference, and stored values are reproduced exactly from checkpoints. They do not establish usefulness.

11. **Main limitations?** Five seeds, 100-epoch checkpoint spacing, coarse temporal control, the shared-difficulty confound, and the unexplained epoch-1,900 onset.

12. **What next, if one wanted to pursue it?** More seeds, denser checkpoints, explain the common onset, control for example difficulty, try other logit reductions. Within this thesis, Higher-MI is closed and Commutator Defect is next.

A Technical appendix

A.1 KSG estimator

All distances use the maximum norm. For point i, ε_i is the distance to its k-th nearest other point in the joint space. Marginal counts are strict: $N_x(i) = \#\{j \neq i : |x_j - x_i| < \varepsilon_i\}$. With ψ the digamma function, T the number of samples and $\langle\cdot\rangle$ the average over points:

$$I(X;Y) = \psi(k) + \psi(T) - \big\langle \psi(N_x+1) + \psi(N_y+1) \big\rangle,$$

$$I(X;Y \mid Z) = \psi(k) + \big\langle \psi(N_z+1) - \psi(N_{xz}+1) - \psi(N_{yz}+1) \big\rangle.$$

Population O-information averages Ω over the six orderings π of (F_0, F_1, F_2): $\Omega_{\text{pop}}(t) = \frac{1}{6}\sum_\pi \big[I(F_{\pi_1};F_{\pi_2}) - I(F_{\pi_1};F_{\pi_2} \mid F_{\pi_3})\big]$. At epoch 12,900 (seed 0) the six-ordering mean is 0.4044 with standard deviation 0.0099, so the estimator's asymmetry is small. F_2 is computed as $(\sum_j \ell[j] - F_0 - F_1)/111$. Mutual information is unchanged by rescaling a variable, so the large logit magnitudes (about 30 to 150) do not affect Ω.

A.2 Frozen protocol

With G_s the grok epoch and P_s the event epoch of seed s, and margin $m = 100$:

$$\text{lead_frac} = \tfrac{1}{5}\sum_s \mathbf{1}[P_s < G_s - 100], \qquad \text{mean_rel_lead} = \tfrac{1}{5}\sum_s \frac{G_s - P_s}{G_s}, \qquad \rho = \text{Spearman}(G, P).$$

PASS requires lead_frac ≥ 0.8, $\rho \geq 0.9$ and $0.05 \leq$ mean_rel_lead ≤ 0.9.

Seed	0	1	2	3	4
Grok epoch G_s	12,987	6,921	9,510	11,019	23,747
Event epoch P_s	15,500	8,500	20,600	12,900	32,500
$\Omega_{\text{pop}}(P_s)$ (nats)	1.2689	1.3089	1.5917	2.3403	2.3347
$(G_s - P_s)/G_s$	−0.1935	−0.2281	−1.1661	−0.1707	−0.3686

Ranks of G: 4,1,2,3,5; ranks of P: 3,1,4,2,5; $\sum d_s^2 = 6$; $\rho = 1 - 6\cdot 6/(5\cdot 24) = 0.7$. (Pearson correlation is 0.8976, but the protocol uses rank correlation.)

Rule validation on synthetic controls (pass rates): ideal $P = 0.5G$: 100%; constant early $P = 5000$: 0%; post-grok $P = 1.5G$: 0%; noisy ideal: 91.04%; random $P \sim U[0, 40000]$: 0.22%; random but always before grok $P \sim U[0, G]$: 21.70%.

A.3 Exploratory signals (reported, not scored)

Signal	P_s, seeds 0–4	lead_frac	ρ	mean_rel_lead
Population zero-crossing	500, 1600, 300, 1100, 200	1.0	−0.7	0.9181
Temporal-control maximum	5000, 5000, 19000, 5000, 14000	0.8	see note	0.1703

The zero-crossing fires very early in every seed. It is the curve wobbling around 0 near initialisation. Its relative lead exceeds 0.9, the same failure as the "constant early" control. The temporal maximum "leads" in three seeds only because epoch 5,000 is the *first* window centre, so the maximum sits at the left edge of the curve. **Flagged discrepancy:** the stored ρ for this row in `aggregate_higher_mi.json` is 0.00, from a rank function that breaks ties by position. The tie-aware Spearman value is 0.11. Either way it is far below 0.9, so no conclusion changes.

Transfer entropy, $TE_{X\to Y}(\tau) = I(Y(t+\tau); X(t) \mid Y(t))$, was computed between the test-mean correct and runner-

up logits for delays $\tau = 1, \ldots, 10$. At $\tau = 1$, the direction correct $\rightarrow$ runner-up is larger in all five seeds. These trajectories change strongly over training, so this is not a clean causal result. It gives no event epoch and is kept only as a record.

A.4 Source of every number and reproduction

All curves and scores come from the project's saved files. The folder is `08_Experiments/results/nanda_unified/`. Inside it:

- `seed_<N>/higher_mi/*.npy` and `higher_mi_signal.json` (curves and events);
- `aggregate_higher_mi.json` (frozen-protocol scores);
- `seed_<N>/training/*_acc_history.npy` (Graph 1).

The single-example logits in Graph 4 are those reported in the source explanation, computed from the seed-0 checkpoints. The full 5-seed recompute took about 27 minutes. Reproduce with:

```
python 06_Code/scripts/run_nanda_benchmark.py \
    --predictors higher_mi --seeds 5 --overwrite higher_mi \
    --output_dir 08_Experiments/results/nanda_unified
python 06_Code/scripts/plot_higher_mi_curves.py
python 06_Code/scripts/analyze_nanda_unified.py
python 06_Code/src/predictors/higher_mi.py   # self-test
```

The shared model file (`transformer_four_head.py`) and the training loop were not changed for this predictor. Rounding note: the thesis wording quotes the mean relative lead as -0.43; the authoritative stored value is -0.4254.